Cyber Anti-Patterns

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Introduction to Antipattern

- An anti-pattern is a common response to a recurring problem that is usually ineffective and risks being highly counterproductive.

Introduction to Antipattern

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Example : Imaginary Prioritization

- So you have sprints, several degrees of priority between P1 and P5, and due dates.
- In practice, due dates are not respected, Px elements maybe completed before or after Px+1 and Px-1, and items are completed sometimes during one sprint, sometimes during one of the following ones.
- So you are prioritizing to death and still cannot forecast when things will be completed.

Introduction to Antipattern

Solution:Use only one method of prioritization and stick to it. If Px prioritization, all P1 must be completed before all P2, and all P2 before all P3.If sprint prioritization, all tasks assigned to an sprint must be completed during the assigned sprint.If using due dates, every item assigned to a due date must be completed by that date, no excuses.

Introduction to Antipattern

- Design forces are the competing concerns, priorities, and technical factors that influence the choice of solutions.
- In antipatterns, there are two solutions: the antipattern solution and the refactored solution.
- An antipattern solution represents a commonplace dysfunctional situation or configuration.
- The antipattern solution may be the result of multiple choices over an extended system lifecycle, or it may have evolved inadvertently.
- Every solution or design choice yields benefits and consequences.
- The refactored solution results from a reconsideration of the design forces and the selection of a more effective solution.
- The refactored solution yields benefits that outweigh its consequences.
- There may also be related solutions or variations that also resolve the design forces beneficially.

Forces in Cyber Antipatterns

The primal design forces in the cybersecurity domain include:

- Management of functionality
- Management of confidentiality
- Management of integrity
- Management of availability

Cyber Antipattern Templates

Micro-Antipattern Templates: It is a flexible and informal way to present antipatterns. The components of a micro-antipattern template are:

- Name: The name of the micro-antipattern is usually a pejorative term, suggesting the negative consequences of the antipattern's presence.
- Antipattern Problem: The problem section summarizes the micro-antipattern's symptoms, consequences, and characterization.
- Refactored Solution: The solution section summarizes alternative ways to resolve the antipattern design forces with improved benefits.

Cyber Antipattern Templates

Full Cyber Antipattern Template: The full cyber antipattern template has two main parts: a header and a body. The header gives a quick sense of the antipattern and the solution, inviting the reader to dive deeper. The heading fields in the full cyber antipattern template are:

- Antipattern Name: The name is a unique pejorative noun phrase. The intent is to make this antipattern a well-known phenomenon, easily recognizable, with an organizational reputation as an important security gap.
- Also Known As: Many antipatterns are known by various names across different organizations. Some known names or analogous names from different domains are listed here. A given organization might want to adopt a name from this list if their members find it suitable.
- Refactored Solution Names: One or more names of alternative solutions are listed here. The purpose is to give the reader a sense of the direction that this pattern write-up is heading toward and to promote a common terminology for the solution identity associated with the antipattern.
- Unbalanced Primal Forces: This field lists the primal design forces that are poorly resolved by this antipattern.
- Anecdotal Evidence: These are some quips that characterize this antipattern. These phrases are sometimes heard when the antipattern is present and in the early recognition of it.

Cyber Antipattern Templates

The body fields in the full cyber antipattern template are:

- Background: This optional field provides contextual explanations that are potentially useful or of general interest but are not central to the antipattern and its refactored solution.
- Antipattern Solution: This field defines the antipattern solution through diagrams, explanations, examples, and discussions of design forces. The antipattern solution is a commonly occurring situation or configuration with significant security implications, such as risks, threats, and vulnerabilities.
- Causes, Symptoms, and Consequences: This bulleted section lists the typical causes, common symptoms, and resulting consequences of the antipattern solution. The intent is to make it easier to recognize the antipattern and understand how and why its replacement is necessary.
- Known Exceptions: If there are some situations where the antipattern solution might be desirable, this section identifies them. For example, if the consequences are acceptable in a context or if replacement is not worthwhile.

Cyber Antipattern Templates

- Refactored Solution and Examples: This field defines the refactored solution. The refactored solution is proposed as an alternative to the antipattern solution. Refactoring is a process of replacing or reworking a given solution into an alternative solution. The new solution resolves the design forces differently, particularly providing a more effective solution that resolves design forces more satisfactorily.
- Related Solutions: If there are other potential solutions to the antipattern, they are identified in this section.
- Often there are different approaches to resolving the same problem that don't conveniently fall under the umbrella of the chosen refactored solution.

Cybersecurity Antipattern Catalog

Can't Patch Dumb

- Antipattern Name: Can't Patch Dumb
- Also Known As: Social Engineering, Phishing, Spam, Spyware, Drive-by Malware, Ransom-Ware, Autoplay Attacks
- Refactored Solution Names: Security Awareness
- Unbalanced Primal Forces: Confidentiality (for example, divulging private information), integrity (for example, rootkits)
- Anecdotal Evidence: "Technology is not the problem; people are the problem," and "Technology is easy; people are difficult."

Cybersecurity Antipattern Catalog

Unpatched Applications

- Antipattern Name: Unpatched Applications
- Also Known As: Vendor-Specific Updates, Default Configuration
- Refactored Solution Names: Patch Management
- Unbalanced Primal Forces: Management of integrity
- Anecdotal Evidence: "Most new attacks are going after the applications, not the operating systems.

Unpatched Applications

Networks Always Play by the Rules

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Unpatched Applications

Never Read the Logs

- Antipattern Name: Never Read the Logs
- Also Known As: Guys Watching Big Network Displays Miss Everything, Insider Threat, Advanced Persistent Threat (APT), Network Operations Center (NOC)
- Refactored Solution Names: Advanced Log Analysis
- Unbalanced Primal Forces: Management of confidentiality
- Anecdotal Evidence: Nick Leeson at Barings Bank, Wikileaks, Aurora Cyber Intrusions

Unpatched Applications

Hard on the Outside, Gooey in the Middle

- Antipattern Name: Hard on the Outside, Gooey in the Middle
- Also Known As: Tootsie Pop, Defense in Depth, Perimeter Security, Protect Everything from All Threats
- Refactored Solution Names: Host Based Security System (HBSS), Network Enclaves
- Unbalanced Primal Forces: Management of confidentiality
- Anecdotal Evidence: "Each user's browser is sending thousands of spyware beacons every day!"; Advanced Persistent Threat; "Our networks are totally secure; we have a firewall."

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Unpatched Applications

Webify Everything

- Antipattern Name: Webify Everything
- Also Known As: Cross-site scripting, Cross-site Request Forgery, US Power Grid on Internet, Global Financial System on Internet
- Refactored Solution Names: Physical Separation, Out of Band Separation
- Unbalanced Primal Forces: Management of integrity and availability
- Anecdotal Evidence: "Why the hell would they put the electrical power grid on the Internet?"

Unpatched Applications

No Time for Security

- Antipattern Name: No Time for Security
- Also Known As: Add Security Last, Blame Security for Schedule Slippage, Deliver It Now!
- Refactored Solution Names: Security Requirements Are Real Requirements, Cyber Risk Management
- Unbalanced Primal Forces: Management of confidentiality, integrity, and availability
- Anecdotal Evidence: "Wait until it's time to test the system, and then worry about security."